Question

When crystal A is heated, it generates gas B, solid C, and water. B reacts with metallic magnesium to produce solid D. D reacts with water to produce gas E and precipitate F. An aqueous solution of A is heated with sodium hydroxide solution to produce gas E and solution G. The addition of barium chloride solution to G produces precipitate H. H is soluble in dilute hydrochloric acid. Given that the thermal decomposition reaction of A is simplified to the lowest integer ratio, and the coefficient of A is 1, what is the mass fraction of hydrogen in crystal A?

A. 1.6%

B. 2.6%

C. 2.9%

D. 3.2%

E. 3.4%

F. 5.0%

G. 5.3%

H. 6.3%

I. 6.4%

J. 6.9%

K. 7.5%

L. 8.4%

M. 6.1%

Answer

Correct Answer: D

Detailed Explanation

According to "Aqueous solution of A reacts with $NaOH$ upon heating to produce gas E and solution G", it is known that when ammonium salts (containing NH_4^+) are heated with strong bases (such as $NaOH$), ammonia gas (NH_3) is produced. Therefore, gas E is likely to be ammonia gas (NH_3).

CHECKPOINT 0.5 PTS

$\mathrm{E} = NH_3$

According to "D reacts with water to produce gas E and precipitate F", and it has already been deduced that gas E is ammonia, then D should be magnesium nitride (Mg_3N_2), and F is magnesium hydroxide ($Mg(OH)_2$).

CHECKPOINT 0.5 PTS

$\mathrm{D} = Mg_3N_2$

According to "B reacts with Mg to produce solid D", and it has already been deduced that D is magnesium nitride (Mg_3N_2), then B may be nitrogen or ammonia. However, since E is ammonia, B is nitrogen (N_2).

CHECKPOINT 0.5 PTS

$B = N_2$

According to "A decomposes upon heating to produce B, C and water", it can be inferred that A is an ammonium salt. Since C is a solid, A should contain a metal acid radical. Ammonium salts of metal acid radicals that come to mind include ammonium chromate, ammonium dichromate, ammonium vanadate, ammonium molybdate, and ammonium tungstate. The following are discussed separately:

Ammonium chromate or ammonium dichromate: G is sodium chromate, H is barium chromate, soluble in dilute hydrochloric acid, which is consistent.

Ammonium vanadate: G is sodium vanadate, H is barium vanadate, soluble in dilute hydrochloric acid, which is consistent.

Ammonium molybdate: G is sodium molybdate, H is barium molybdate, and the molybdic acid produced by the reaction with dilute hydrochloric acid is slightly soluble, which is inconsistent.

Ammonium tungstate: G is sodium tungstate, H is barium tungstate, and the tungstic acid produced by the reaction with dilute hydrochloric acid is insoluble, which is inconsistent.

Possible A is ammonium chromate, ammonium dichromate, ammonium vanadate, among which only the simplest formula of the thermal decomposition reaction of ammonium dichromate has a coefficient of 1 for A. Therefore, A is ammonium dichromate ($(NH_4)_2Cr_2O_7$).

CHECKPOINT 1 PTS

$A = (NH_4)_2Cr_2O_7$

The mass fraction of H is calculated to be 3.199%, choose D.

CHECKPOINT 0.5 PTS

$w_{\mathrm{H}} = 3.199\ \%$